

How difficult is it to derive a stock specific market impact model? In Chapter 5 we discussed techniques to derive a general equation market impact formula and a universal set of parameters across all stocks (a_1, a_2, a_3, a_4 , and b_1). Since this model incorporates stock volatility and stock ADV into the formulation it in a way provides different estimates across stocks. But recall the last part of the chapter where we uncovered a relationship between stock errors and company fundamental data. We found strong evidence that market impact is negatively correlated with log of market cap and log of price, and positively correlated with beta, spreads, and stock specific risk (tracking error). This finding suggests that we might still be able to improve our results at the stock level by including these data. But then why not simply calibrate a stock specific model similar to how we have stock specific volatility and beta estimates. The short answer is that we simply do not have enough data available. Price movement is often dominated by market movement, market noise (volatility), and buying and selling pressure from all other investors. This makes it very difficult to uncover stable statistical relationships between price movement and customer orders.

To illustrate the difficulty associated with estimating stock specific parameters let us examine the estimation process using simulated data¹. We know that one of the biggest drawbacks with market impact estimation is the “Heisenberg uncertainty principle of trading”; that is, we can only observe the price trajectory with the order or the price trajectory without the order—not both. Therefore, we are not able to accurately determine price movement caused solely by the order. Well this is certainly true in reality. But let’s take a step back for a moment. Suppose that we do know the exact relationship between price movement and buying and selling pressure of a trade. Then we can simulate trade data and test our market impact estimation approach on the simulated data to determine if we have an accurate estimation technique. The important point to keep in mind is that stock price is driven by many factors such as stock specific alpha, general market movement, impact from the order, buying and selling pressure from other participants, and price volatility. If we know exactly the relationship between price impact and buying-selling pressure we can simulate a market impact data series, and then test our model against that simulated series to determine if our modeling approach is able to uncover the true relationship.

¹A variation of this exercise was previously given to my Cornell University Financial Engineering graduate students as a final project, and it does a great job to highlight the difficulty in constructing stock specific parameters.